9th Class • Punjab Board • PECTAA New Syllabus

Chapter 3: Dynamics

Complete solved exercise — Newton’s laws, momentum, impulse, friction, collisions and recoil numericals.

Chapter 3 • Complete official exercise • 9 MCQs + 10 short + 6 constructed + 6 comprehensive + 9 numericals

What this chapter is about

• **Force** is an interaction that can change velocity or deform a body. Net force, not each individual force, determines acceleration.
• **Momentum** p = mv is a vector. Impulse FΔt = Δp explains catching, collisions and safety devices.
• **Newton’s third law** concerns forces on two different bodies. Internal action–reaction pairs conserve total momentum in an isolated system.
• **Friction** can oppose relative motion or provide essential grip. Static friction adjusts up to a limit; kinetic friction acts during sliding.

Sign rule: Choose a positive direction before every momentum calculation. Use negative velocities for motion in the opposite direction. Take g = 10 m s^{-2} for the numericals.

A. Multiple Choice Questions

1. When we kick a stone, we get hurt. This is due to:

(a) inertia
(b) velocity
(c) momentum
(d) reaction

Answer: (d) reaction — The stone exerts a reaction force on the foot equal in magnitude and opposite in direction to the foot’s force on the stone.

2. An object will continue its motion with constant acceleration until:

(a) the resultant force on it begins to decrease.
(b) the resultant force on it is zero.
(c) the resultant force on it begins to increase.
(d) the resultant force is at right angle to its tangential velocity.

Answer: (a) the resultant force on it begins to decrease. — For constant mass a = F/m, so constant acceleration requires a constant resultant force. Option (a) is the intended choice. Strictly, an increase also changes acceleration, while zero resultant force makes acceleration zero; the wording is therefore imprecise.

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3. Which of the following is a non-contact force?

(a) Friction (b) Air resistance
(c) Electrostatic force (d) Tension in the string

Answer: (c) Electrostatic force — Electrostatic force acts between charges without material contact. Friction, air resistance and tension require contact with a surface, fluid or string.

4. A ball with initial momentum p hits a solid wall and bounces back with the same velocity. Its momentum p′ after collision will be:

(a) p′ = p (b) p′ = −p
(c) p′ = 2p (d) p′ = −2p

Answer: (b) p′ = −p — Here ‘same velocity’ must mean same speed but reversed direction. Momentum is a vector, so the rebound momentum is −p.

5. A particle of mass m moving with a velocity v collides with another particle of the same mass at rest. The velocity of the first particle after collision is:

(a) v (b) −v
(c) 0 (d) −v/2

Answer: (c) 0 — For a head-on elastic collision between equal masses, the incoming particle stops and transfers its velocity to the particle initially at rest.

6. Conservation of linear momentum is equivalent to:

(a) Newton’s first law of motion (b) Newton’s second law of motion
(c) Newton’s third law of motion (d) None of these

Answer: (c) Newton’s third law of motion — Equal and opposite interaction forces acting for the same time produce equal and opposite momentum changes, leaving total momentum unchanged.

7. An object with a mass of 5 kg moves at constant velocity of 10 m s^{-1}. A constant force then acts for 5 seconds and gives it a velocity of 2 m s^{-1} in the opposite direction. The force acting on the object is:

(a) 5 N (b) −10 N
(c) −12 N (d) −15 N

Answer: (c) −12 N — Take the initial direction as positive. F = m(v − u)/t = 5(−2 − 10)/5 = −12 N; the force opposes the initial motion.

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8. A large force acts on an object for a very short interval of time. In this case, it is easy to determine:

(a) magnitude of force
(b) time interval
(c) product of force and time
(d) none of these

Answer: (c) product of force and time — The measurable quantity is impulse $J = F\Delta t = \Delta p$, obtained from the momentum change even when force and contact time are difficult to measure separately.

9. A lubricant is usually introduced between two surfaces to decrease friction. The lubricant:

(a) decreases temperature
(b) acts as ball bearings
(c) prevents direct contact of the surfaces
(d) provides rolling friction

Answer: (c) prevents direct contact of the surfaces — A lubricant forms a film that separates surface irregularities and reduces direct solid-to-solid contact.

B. Short Answer Questions

3.1. What kind of changes in motion may be produced by a force?

A net force can start or stop motion, increase or decrease speed, change direction, or produce a combination of these changes. A force can also deform a body without changing its overall motion.

3.2. Give 5 examples of contact forces.

- Normal reaction between a book and table
- Friction between shoes and floor
- Tension in a stretched string
- Air resistance on a moving ball
- Muscular push or pull on a cart

3.3. An object moves with constant velocity in free space. How long will the object continue to move with this velocity?

Indefinitely, as long as no net external force acts on it. By Newton’s first law, its speed and direction remain unchanged in the absence of a resultant force.

3.4. Define impulse of force.

Impulse is the product of a force and the time interval for which it acts: $J = F\Delta t$ for constant force. It is a vector measured in N s and equals the change in momentum, $J = \Delta p$.

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3.5. Why has not Newton’s first law been proved on the Earth?

Its ideal condition—zero net external force—cannot be perfectly achieved on Earth because gravity, friction and air resistance are present. Experiments can reduce and balance these forces and approach the law closely, but cannot provide a perfectly force-free demonstration.

3.6. When sitting in a car which suddenly accelerates from rest, you are pushed back into the seat, why?

Your body tends to remain at rest because of inertia while the seat moves forward. The seat back then exerts the forward force needed to accelerate you. The feeling of being pushed backward is relative to the accelerating car, not a separate backward force in an inertial frame.

3.7. The force expressed in Newton’s second law is a net force. Why is it so?

Several forces may act simultaneously and their vector sum determines acceleration. Balanced components cancel, so only the unbalanced resultant changes momentum: $F_{net} = \Delta p/\Delta t = ma$ for constant mass.

3.8. How can you show that rolling friction is lesser than the sliding friction?

Place the same loaded object on a horizontal surface and measure the force needed to keep it moving with a spring balance, first while sliding and then while rolling on rollers or wheels. For the same normal load, the smaller steady pulling force in the rolling case shows that rolling friction is smaller.

3.9. Define terminal velocity of an object.

Terminal velocity is the constant maximum velocity reached by a body falling through a fluid when downward weight is balanced by upward drag and buoyancy. The resultant force and acceleration then become zero.

3.10. An astronaut walking in space wants to return to his spaceship by firing a hand rocket. In what direction does he fire the rocket?

Direct the rocket exhaust **away from the spaceship**, opposite to the desired motion. The backward ejected mass gains momentum, so conservation of momentum gives the astronaut an equal forward momentum toward the spaceship.

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C. Constructed Response Questions

3.1. Two ice skaters weighing 60 kg and 40 kg push off against each other on a frictionless ice track. The 60 kg skater gains a velocity of 4 m s^{-1}. Considering all the relevant calculations involved, explain how Newton's third law applies to this situation.

Each skater exerts an equal-magnitude, opposite-direction force on the other for the same contact time. Their impulses and momentum changes are therefore equal and opposite. Initially the total momentum is zero.

$$m_1v_1 + m_2v_2 = 0$$

$$60 \times 4 + 40 \times v_2 = 0$$

$$v_2 = -240/40 = -6 \text{ m s}^{-1}$$

The 40 kg skater moves at **6 m s^{-1} opposite** to the 60 kg skater. The interaction forces are equal, but the smaller mass receives the larger speed and acceleration.

3.2. Inflatable airbags are installed in vehicles as safety equipment. In terms of momentum, what is the advantage of airbags over seatbelts?

The passenger's momentum change during a crash is fixed, Δp = F_avgΔt. An airbag increases stopping time and spreads the force over a larger area, reducing average force and pressure on the head and chest. A seatbelt also increases stopping time, but its force is concentrated over straps; the two systems work best together.

3.3. A horse refuses to pull a cart, arguing that the cart exerts an equal and opposite force, so the net force is zero and it cannot accelerate. What is wrong with this argument?

The horse's force on the cart and cart's force on the horse act on **different bodies**, so they do not cancel in either body's free-body diagram. The ground exerts forward friction on the horse; the horse pulls the cart forward. If the forward force on the horse–cart system exceeds resisting friction, the whole system accelerates.

3.4. When a cricket ball hits high, a fielder tries to catch it. While holding the ball he/she draws hands backward. Why?

Drawing the hands backward lengthens the time over which the ball's momentum becomes zero. Since F_avg = Δp/Δt, a larger Δt gives a smaller average force on the hands, making the catch safer and less likely to be dropped.

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3.5. When someone jumps from a small boat onto the river bank, why does the jumper often fall into the water? Explain.

To jump forward, the person pushes backward on the boat. The light boat moves backward because momentum is conserved and water offers limited resistance. The foot therefore loses support and the effective forward push or reach is reduced, so the jumper may not reach the bank and can fall into the water.

3.6. Imagine that if friction vanishes suddenly from everything, then what could be the scenario of daily life activities?

Walking, running and vehicle starting or braking would become impossible; tyres and shoes would slip. We could not grip pens, cups or tools, knots would untie, nails and screws would not hold, and objects on slopes would slide. Machinery would lose less energy to heat and wear, but controlled motion and ordinary gripping would collapse.

D. Comprehensive Questions

3.1. Explain the concept of force by practical examples.

A **force** is a push or pull arising from interaction between bodies. It is a vector measured in newtons (N), where 1 N = 1 kg m s^{-2}. A net force changes momentum; forces can also deform objects.

Situation	Effect of force
Kick a stationary football	Changes speed from zero and sets direction.
Apply brakes to a bicycle	Friction decreases speed.
Hit a ball with a bat	Changes momentum and direction in a short time.
Stretch a spring or squeeze dough	Changes shape or size.
Earth attracts a falling stone	Non-contact gravitational force produces downward acceleration.

Contact forces require physical contact, while gravitational, electrostatic and magnetic forces can act at a distance. The acceleration depends on the **net external force**.

3.2. Describe Newton’s laws of motion.

1. First law (inertia): A body remains at rest or in uniform straight-line motion unless acted upon by a net external force. Inertia is the tendency to resist change in velocity; mass measures it.

2. Second law: The rate of change of momentum equals the net external force and occurs in its direction: F_net = Δp/Δt. For constant mass, F_net = ma, so a = F_net/m.

3. Third law: When body A exerts a force on B, B simultaneously exerts an equal-magnitude, opposite-direction force on A. The forces have the same nature, act along the same line and act on different bodies.

Examples: passengers lurch when a bus starts; a smaller mass accelerates more for the same force; a rocket moves forward by expelling gas backward.

3.3. Define momentum and express Newton's second law of motion in terms of change in momentum.

Linear momentum is the product of mass and velocity: $p = mv$. It is a vector in the direction of velocity and its SI unit is kg m s^{-1} (equivalent to N s).

Initial momentum $p_i = mu$; final momentum $p^f = mv$

$$F_{net} = (p^f - p_i)/\Delta t = \Delta p/\Delta t$$

For constant mass: $F_{net} = m(v - u)/\Delta t = ma$

Impulse: $J = F_{net}\Delta t = \Delta p$

The equation also handles changing mass conceptually and shows why increasing impact time reduces average force for a fixed momentum change.

3.4. State and explain the principle of conservation of momentum.

Principle: If no net external force acts on a system, its total linear momentum remains constant, although individual momenta may change.

$$p_{total,before} = p_{total,after}$$

For two bodies: $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$

By Newton's third law, internal forces are equal and opposite and act for the same time. Therefore $\Delta p_1 = F_{12}\Delta t$ and $\Delta p_2 = -F_{12}\Delta t$, so $\Delta(p_1 + p_2) = 0$.

A fired gun recoils because the forward momentum of the bullet is balanced by backward momentum of the gun. The principle applies to explosions and collisions when external impulse is negligible.

3.5. Describe the motion of a block on a table taking into account the friction between the two surfaces. What is static friction and kinetic friction?

• **No applied horizontal force:** no horizontal friction is needed and the block remains at rest.

• **Small applied force:** static friction adjusts equally and oppositely, so the block remains at rest until the limiting static friction is reached.

• **Force greater than limiting static friction:** the block starts sliding. Kinetic friction opposes relative sliding; acceleration $a = (F_{applied} - F_k)/m$.

• **Constant velocity:** applied force equals kinetic friction and net force is zero. If the applied force is removed, kinetic friction decelerates the block to rest.

Static friction acts between surfaces with no relative sliding and varies up to a maximum. **Kinetic (sliding) friction** acts when surfaces slide over each other; for a given normal load it is usually smaller than limiting static friction.

3.6. Explain the effect of friction on the motion of vehicles in context of tyre surface and braking force.

Tyre–road friction provides the external force needed to start, steer and stop a vehicle. During normal rolling without skidding, the contact patch is instantaneously at rest relative to the road and static friction provides grip.

• **Braking:** Brake friction slows the wheel, while tyre–road friction supplies the backward force on the vehicle. Maximum safe braking uses available static friction without locking the wheels.

• **Locked wheels:** Sliding kinetic friction replaces static grip, steering control decreases and stopping distance often increases. Anti-lock braking systems reduce locking.

• **Tread and surface:** Good tread and a dry rough road provide better grip. Bald tyres, water, oil, sand or ice reduce available friction, lengthening stopping distance.

• **Load and pressure:** Correct tyre pressure and sufficient contact help predictable grip; excessive speed still demands more friction than a bend or stop can safely provide.

E. Numerical Problems

3.1 A 10 kg block is placed on a smooth horizontal surface. A horizontal force of 5 N is applied to the block. Find (a) the acceleration produced in the block (b) the velocity of block after 5 seconds.

Given: m = 10 kg; F = 5 N; u = 0; t = 5 s. Smooth means friction is negligible.

$$a = F/m = 5/10 = 0.5 \text{ m s}^{-2}$$

$$v = u + at = 0 + 0.5 \times 5 = 2.5 \text{ m s}^{-1}$$

Answer: (a) 0.5 m s⁻² (b) 2.5 m s⁻¹.

3.2 The mass of a person is 80 kg. What will be his weight on the Earth? What will be his weight on the Moon? The value of acceleration due to gravity of Moon is 1.6 m s⁻².

Given: m = 80 kg; g_E = 10 m s⁻²; g_M = 1.6 m s⁻². Mass stays 80 kg.

$$W_E = mg_E = 80 \times 10 = 800 \text{ N}$$

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$$W_M = mg_M = 80 \times 1.6 = 128 \text{ N}$$

Answer: Earth weight = 800 N; Moon weight = 128 N.

3.3 What force is required to increase the velocity of 800 kg car from 10 m s^{-1} to 30 m s^{-1} in 10 seconds?

Given: m = 800 kg; u = 10 m s^{-1}; v = 30 m s^{-1}; t = 10 s.

$$a = (v - u)/t = (30 - 10)/10 = 2 \text{ m s}^{-2}$$

$$F = ma = 800 \times 2 = 1600 \text{ N}$$

Answer: 1600 N in the direction of acceleration.

3.4 A 5 g bullet is fired by a gun. The bullet moves with a velocity of 300 m s^{-1}. If the mass of the gun is 10 kg, find the recoil speed of the gun.

Given: m_b = 0.005 kg; v_b = +300 m s^{-1}; m_g = 10 kg; initial momentum = 0.

$$0 = m_bv_b + m_gv_g$$

$$v_g = -(0.005 \times 300)/10 = -1.5/10 = -0.15 \text{ m s}^{-1}$$

Answer: Recoil speed = 0.15 m s^{-1}; recoil velocity = −0.15 m s^{-1} (opposite to bullet).

3.5 An astronaut of mass 70 kg throws a wrench of mass 300 g at a speed of 3.5 m s^{-1}. Determine (a) the speed of astronaut as he recoils away from the wrench (b) the distance covered by the astronaut in 30 minutes.

Given: m_A = 70 kg; m_w = 0.300 kg; v_w = +3.5 m s^{-1}; t = 30 min = 1800 s.

$$0 = m_Av_A + m_wv_w$$

$$v_A = -(0.300 \times 3.5)/70 = -0.015 \text{ m s}^{-1} = -1.5 \times 10^{-2} \text{ m s}^{-1}$$

$$\text{Distance} = \text{speed} \times \text{time} = 0.015 \times 1800 = 27 \text{ m}$$

Answer: (a) 0.015 m s^{-1} opposite to wrench (b) 27 m.

3.6 A 6.5×10^3 kg bogie moving at 0.8 m s^{-1} is struck from behind by a 9.2×10^3 kg bogie moving at 1.2 m s^{-1}. They couple. Find their common velocity.

Given: m_1 = 6500 kg; u_1 = 0.8 m s^{-1}; m_2 = 9200 kg; u_2 = 1.2 m s^{-1}. Same positive direction.

$$(m_1 + m_2)v = m_1u_1 + m_2u_2$$

$$v = (6500 \times 0.8 + 9200 \times 1.2)/(6500 + 9200)$$

$$v = (5200 + 11040)/15700 = 16240/15700 = 1.034 \text{ m s}^{-1}$$

Answer: Common velocity ≈ 1.03 m s⁻¹ forward.

3.7 A cyclist of mass 55 kg rides a bicycle of mass 5 kg. He starts from rest and applies a force of 90 N for 8 seconds. Then he continues at a constant speed for another 8 seconds. Calculate total distance travelled.

Given: Total mass = 60 kg; F = 90 N; u = 0; $t_1 = t_2 = 8$ s.

$$a = F/m = 90/60 = 1.5 \text{ m s}^{-2}$$

After 8 s: $v = at_1 = 1.5 \times 8 = 12 \text{ m s}^{-1}$

$$S_1 = \tfrac{1}{2}at_1^2 = \tfrac{1}{2} \times 1.5 \times 8^2 = 48 \text{ m}$$

$$S_2 = vt_2 = 12 \times 8 = 96 \text{ m}$$

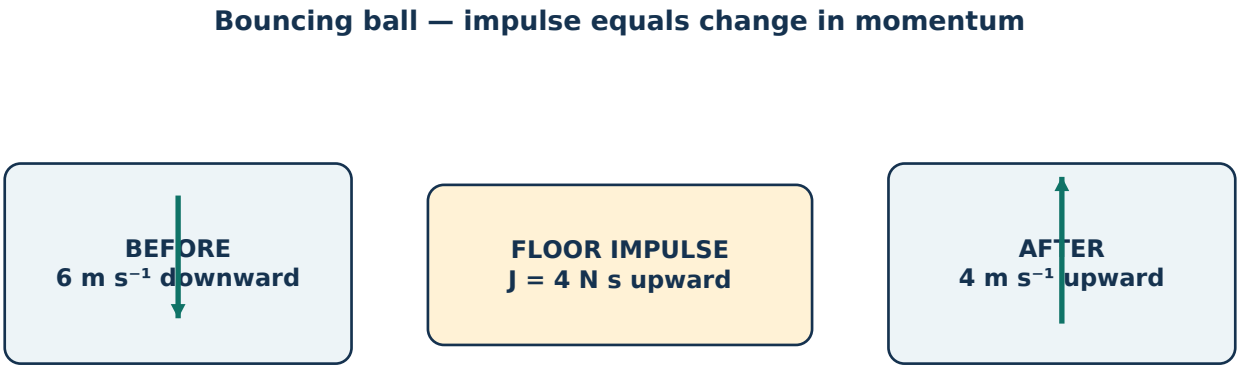
Total S = 48 + 96 = 144 m

Answer: 144 m.

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3.8. A ball of mass 0.4 kg is dropped on the floor from a height of 1.8 m. The ball rebounds straight upward to a height of 0.8 m. What is the magnitude and direction of the impulse applied to the ball by the floor?

Figure 1 — Velocities immediately before and after floor contact



Take upward as positive: $p_i = 0.4(-6) = -2.4$ kg m s⁻¹

$p^f = 0.4(+4) = +1.6$; $J = p^f - p_i = 1.6 - (-2.4) = +4.0$ N s

Positive sign means that the floor's impulse on the ball is upward.

Impact speed = $\sqrt{(2gh_1)} = \sqrt{(2 \times 10 \times 1.8)} = 6$ m s^{-1} downward

Rebound speed = $\sqrt{(2gh_2)} = \sqrt{(2 \times 10 \times 0.8)} = 4$ m s^{-1} upward

Take upward positive: $u = -6$ m s^{-1} and $v = +4$ m s^{-1}

$J = m(v - u) = 0.4[4 - (-6)] = +4$ N s

Answer: 4 N s upward. The large sign change occurs because momentum reverses direction.

3.9 Two balls of masses 0.2 kg and 0.4 kg move towards each other with velocities 20 m s⁻¹ and 5 m s⁻¹ respectively. After collision, the velocity of the 0.2 kg ball becomes 6 m s⁻¹. What will be the velocity of the 0.4 kg ball?

Given: Take the 0.2 kg ball's initial direction positive. $u_1 = +20$; $u_2 = -5$; final $v_1 = +6$ m s^{-1} as used by the stated exercise answer.

$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$

$0.2(20) + 0.4(-5) = 0.2(6) + 0.4v_2$

$4 - 2 = 1.2 + 0.4v_2$

$v_2 = (2 - 1.2)/0.4 = +2$ m s^{-1}

Answer: 2 m s⁻¹ in the original direction of the 0.2 kg ball.

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F. More Important Questions (from the text)

1. Why are action and reaction forces unable to cancel each other?

Action and reaction are equal in magnitude and opposite in direction, but they act on **different bodies**. Forces can be added or cancelled only when they act on the same body or system.

For example, Earth pulls an apple downward and the apple pulls Earth upward. The apple accelerates noticeably because of its small mass; Earth's acceleration is extremely small. The two forces do not produce zero net force on the apple.

2. Distinguish clearly between mass and weight.

Mass	Weight
Amount of matter and measure of inertia.	Gravitational force on a body.
Scalar; SI unit kilogram (kg).	Vector toward the attracting body; SI unit newton (N).
Measured with a balance by comparison.	Measured with a calibrated spring balance or calculated as W = mg.
Remains constant at a given location in classical physics.	Changes with local gravitational acceleration; nearly zero in free-fall weightlessness.

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Points to Remember

Force is an interaction; acceleration is produced by net external force.

Inertia: a body preserves rest or uniform straight-line motion when F_net = 0.

Newton II: F_net = Δp/Δt; for constant mass F_net = ma.

Newton III: action and reaction are equal, opposite, simultaneous and act on different bodies.

Momentum: p = mv; unit kg m s^{-1} = N s; direction is velocity direction.

Impulse: J = FΔt = Δp; increasing impact time reduces average force.

Conservation: $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$ when external impulse is negligible.

Weight: W = mg; mass is unchanged from Earth to Moon.

Static friction adjusts until its limit; kinetic friction acts during sliding.

Rolling friction is generally smaller than sliding friction for the same load.

MCQ key at a glance

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	d	2	a	3	c	4	b	5	c
6	c	7	c	8	c	9	c		

How to attempt this chapter in the paper

• Draw a direction arrow and assign signs before substituting into momentum equations.
• For recoil from rest, total initial momentum is zero; one recoil velocity must be negative.
• For a bounce, final minus initial velocity includes the reversal; do not subtract only speeds.
• For coupled objects, use combined mass after finding total initial momentum.
• In force–time safety questions, keep Δp fixed and explain how larger Δt lowers F_avg.
• Never cancel an action–reaction pair on one free-body diagram: the forces act on different bodies.